



Track Number: 3

**Track Title: Technology for Social Good &
Sustainable Progress**

AI-Driven Integrated Urban Resilience System

Video Link: (Optional)

Team ID: ID_5_N+1's

Team Name: N+1's

Institute Name: LDRP-ITR



TEAM DETAILS



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IDEA DETAILS



IDEA

We propose a unified, AI-powered civic platform where citizens can easily report road and public safety issues using real-time photos and GPS. These reports are intelligently verified, prioritized, and displayed on a single dashboard, enabling authorities to take faster, more coordinated action and ensure safer urban spaces.



KEY PROBLEM

- Urban civic services operate on multiple disconnected (siloed) platforms
- Citizens must use different apps/portals to report basic road and safety issues
- Delayed response and poor coordination among departments
- Lack of real-time, location-verified data leads to unsafe roads and public spaces



SOLUTION

- One unified, AI-powered platform to manage all road and public safety issues
- Easy citizen reporting using real-time photos/videos with automatic GPS location
- AI-based verification to ensure genuine reports, remove duplicates, and prioritize issues
- Transparent issue lifecycle: Submitted → Verified → Assigned → In Progress → Resolved
- Centralized authority dashboard with map-based insights for faster action

TECHNICAL APPROACH



FRONTEND (USER INTERFACE)



HTML5-STRUCTURE OF WEB APPLICATION



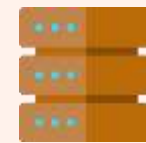
CSS3-STYLING AND LAYOUT (tailwind.css, etc.)



JAVA SCRIPT-INTERACTIVE USER EXPERIENCE



CAMERA&GPS API-IMAGE CAPTURE AND LOCATION



BACKEND (SERVER SIDE)



NODE.JS-SERVER-SIDE RUNTIME ENVIRONMENT



EXPRESS.JS -API HANDLING AND ROUTING



REST API'S- COMMUNICATION BETWEEN FRONTEND & BACKEND



AUTHENTICATION MODULE-SECURE USER AUTHENTICITY ACCESS



DATABASE LAYER



MONGO DB

- STORES COMPLAINT DATA
- IMAGE METADATA
- GPS LOCATION



MY SQL

- USER PROFILES
- AUTHORITY AND OFFICER DETAILS
- LOGIN AUTHENTICATION



AI/PROCESSING LAYER



JAVA SCRIPT BASED AI INTEGRATION



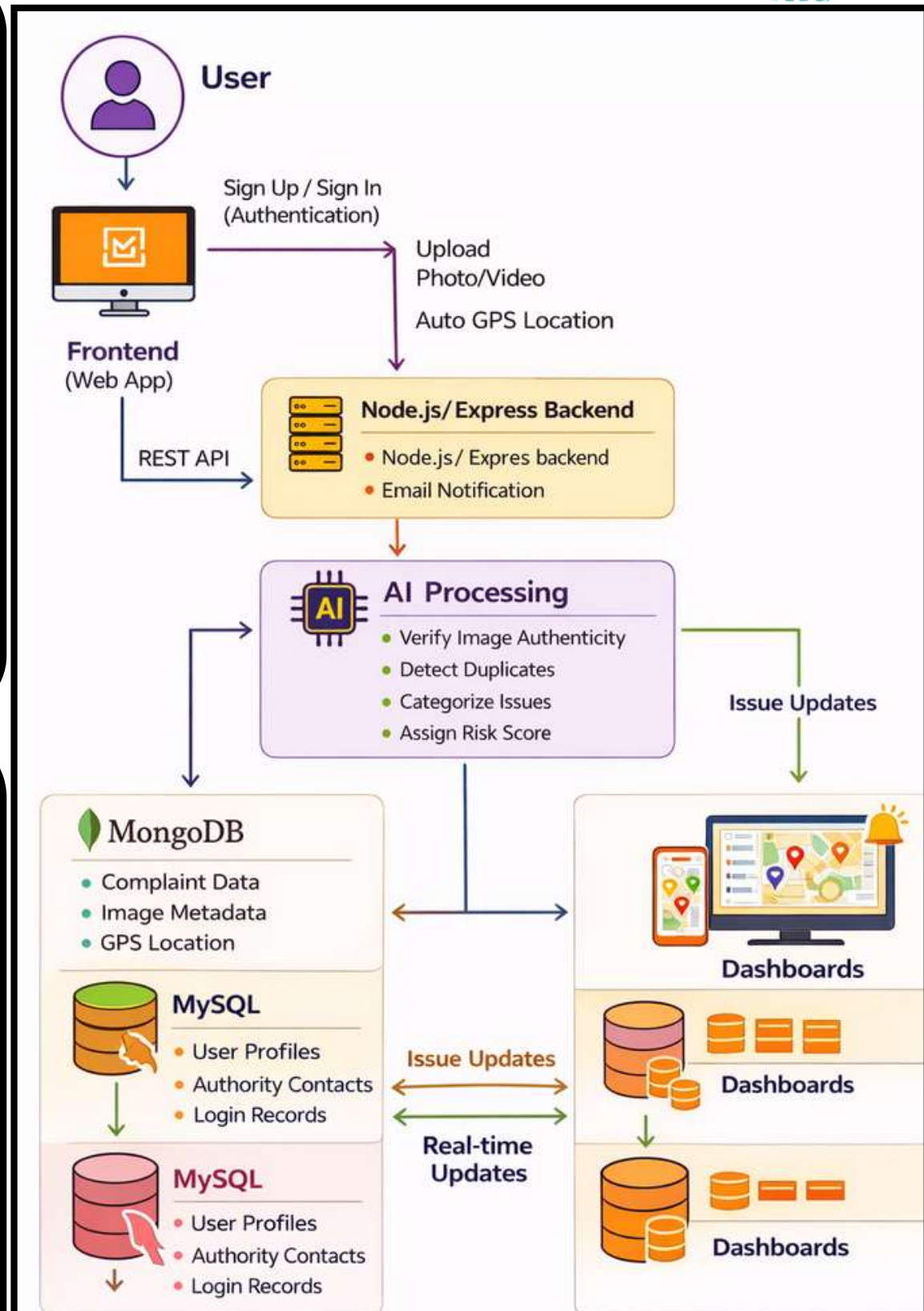
IMAGE AUTHENTICITY VERIFICATION



DUPLICATE IMAGE & FALSE LOCATION DETECTION



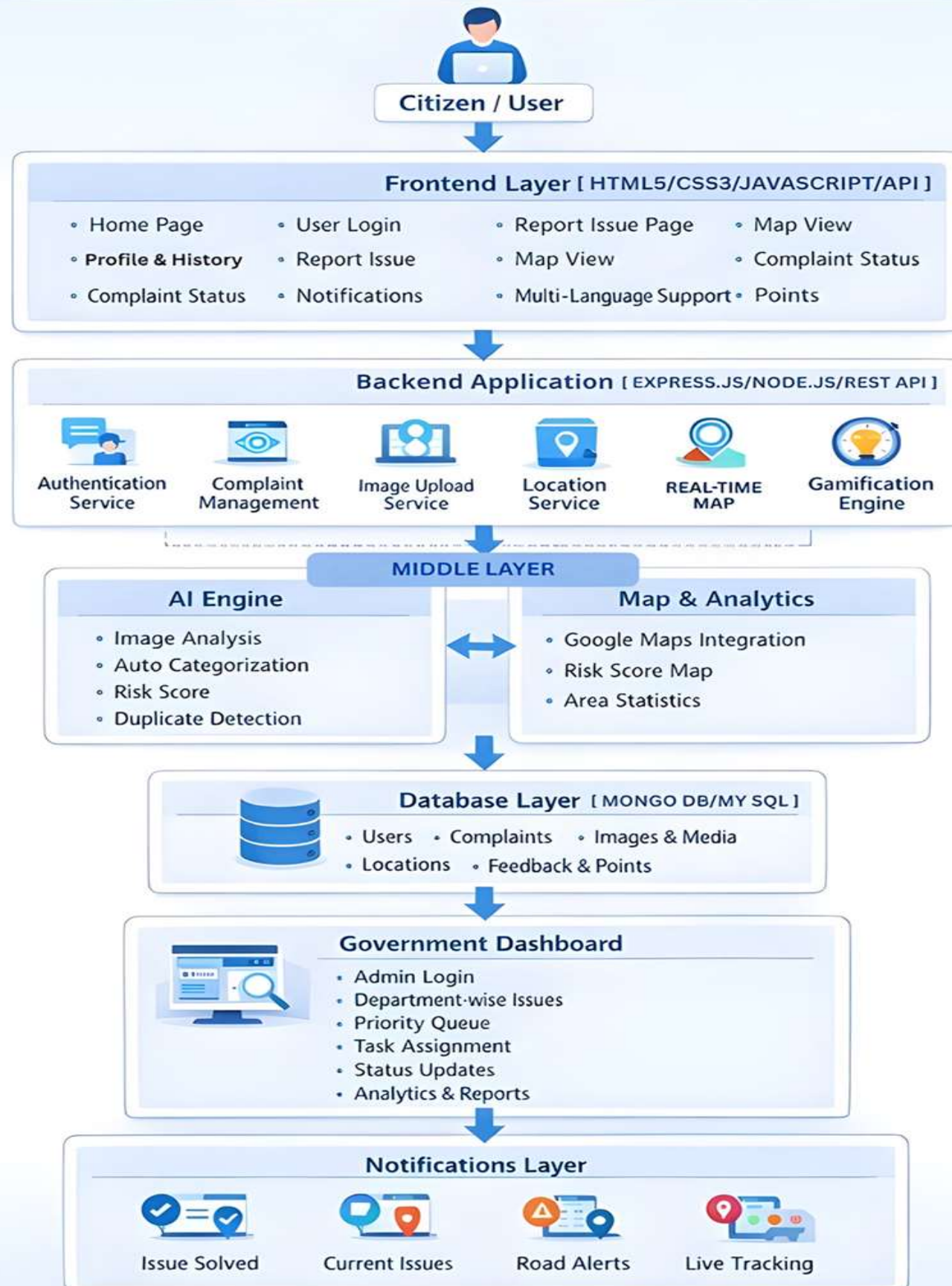
RISK SCORE CALCULATION



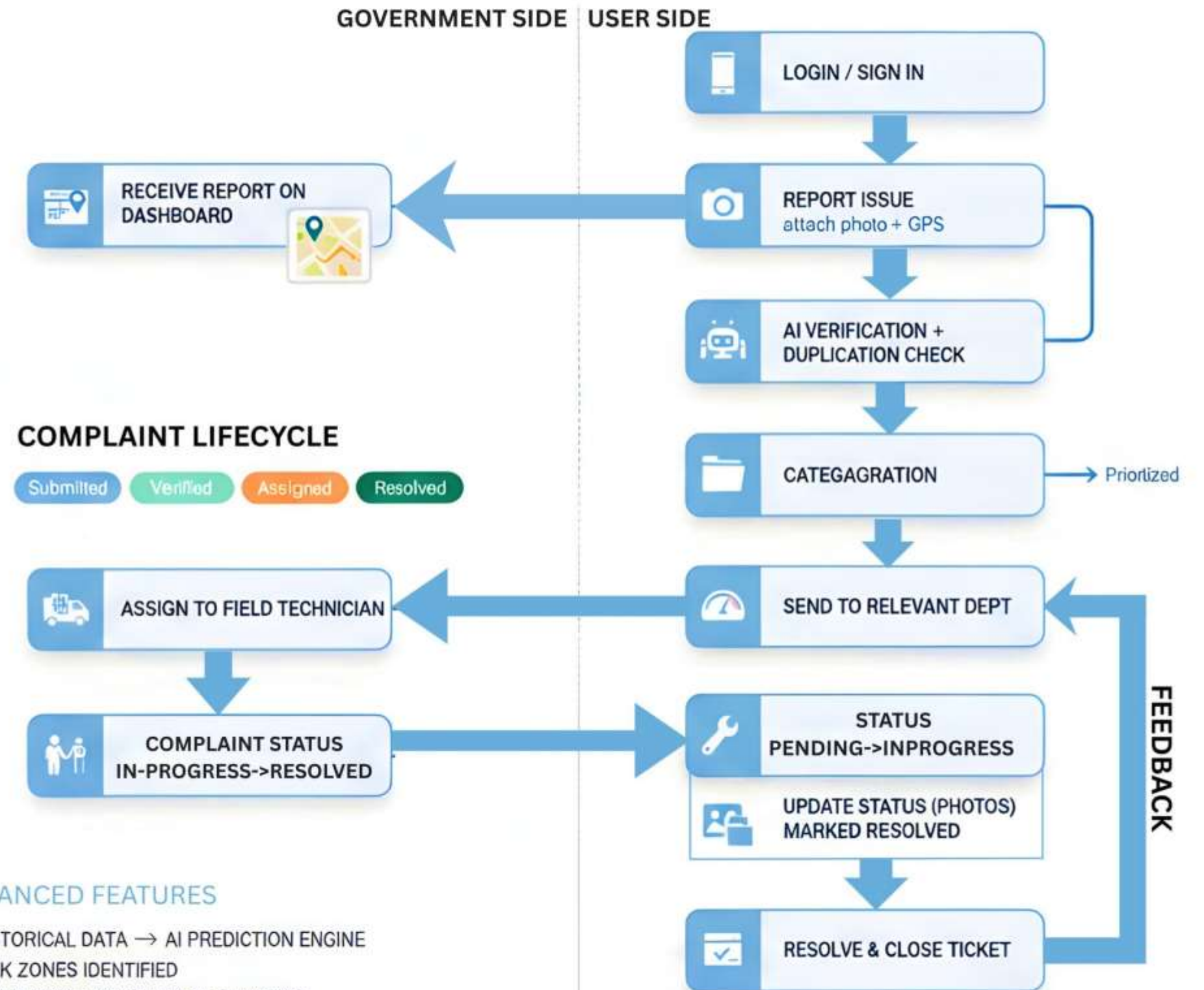
ARCHITECTURE



Architecture Flowchart



WEBSITE WORKFLOW



FEASIBILITY AND VIABILITY



Feasibility

- Uses existing web, AI, and map technologies
- Works with camera & GPS already available on smartphones
- Scalable architecture (Ward → City → State)



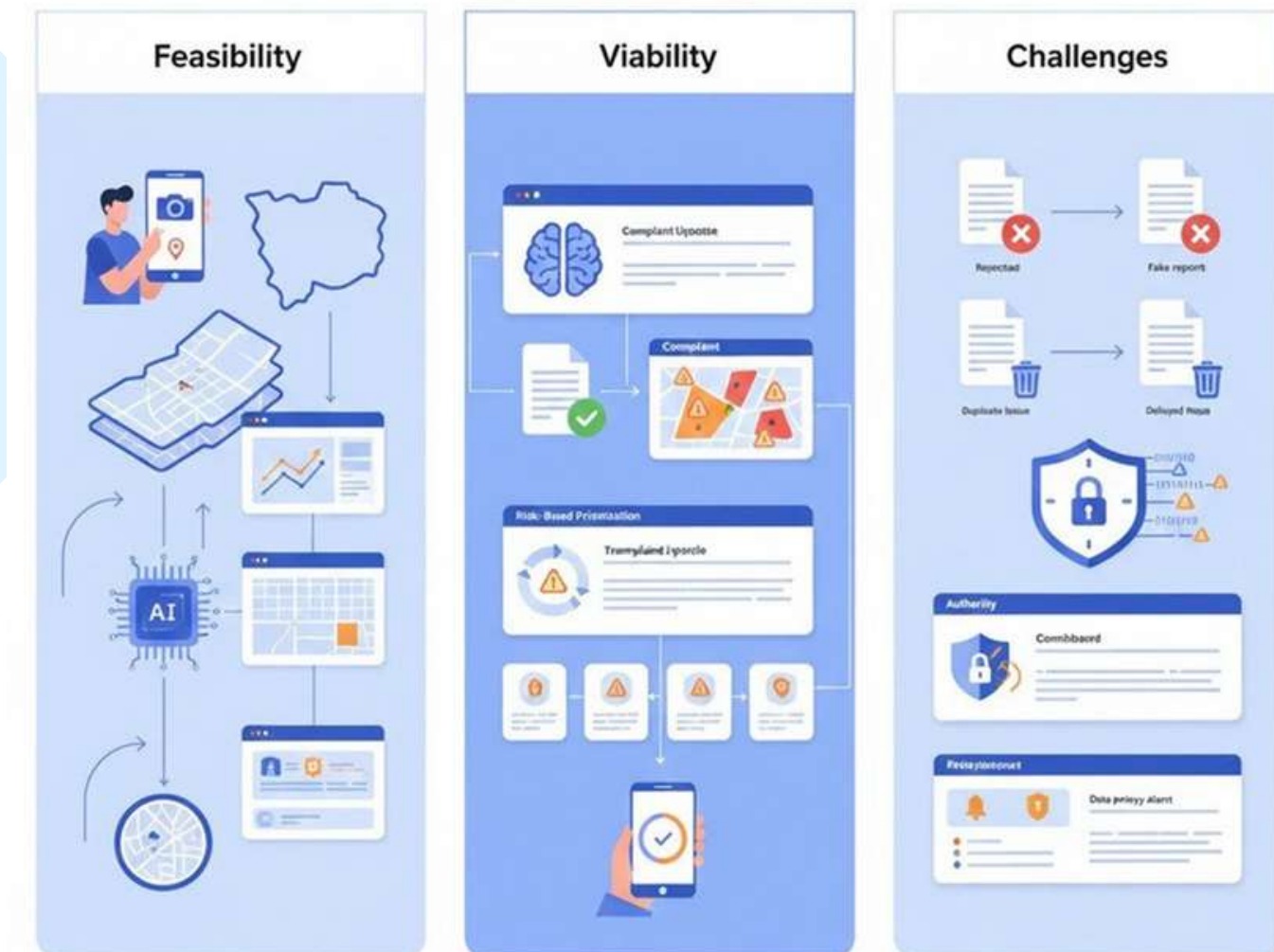
Challenges

- Fake or duplicate issue reports
- Delays in authority response
- Data privacy & user trust



Viability

- AI verification & duplicate detection
- Risk-based prioritization for faster action
- Transparent tracking builds citizen trust



IMPACT AND BENEFITS



TECHNICAL EDGE

- The AI Advantage
- AI verifies real damage and filters fake or generated images
- Automated risk scoring prioritizes high-traffic danger zones
- Multi-language support ensures inclusivity for all citizens



GOVERNMENT EFFICIENCY

- Smart Governance
- Much faster than manual road audits, saving time and manpower
- Prevents budget waste through duplicate complaint detection
- Ensures full transparency with live tracking:
- Submitted → Assigned → Resolved



SOCIAL IMPACT

- Saving Lives, Not Just Roads
- Prevents two-wheeler accidents by detecting potholes and unsafe roads early
- Improves pedestrian safety by flagging broken streetlights and blocked pathways
- Predicts accident-prone black spots using historical data for preventive action



<u>FEATURE / CRITERIA</u>	<u>PROPOSED UNIFIED CIVIC PLATFORM</u>	<u>EXISTING DEPARTMENT PORTALS</u>	<u>MANUAL / HELPLINE-BASED SYSTEM</u>
SINGLE UNIFIEDPLATFORM	Yes – all departments integrated	No – separate portals per department	No unified system
ISSUECLASSIFICATION	Automated using AI and rules	Manual category selection	Manual
RISK-BASED PRIORITIZATION	Built-in hazard severity scoring	Not available	Not available
CROSS-DEPARTMENT COORDINATION	Centralized workflow management	Limited or absent	Not available
REAL-TIME STATUS TRACKING	End-to-end tracking and notification	Partial visibility	Not supported
DUPLICATE COMPLAINT DETECTION	Automated detection	Not supported	Not available
ANALYTICS & HAZARD HEATMAPS	State-level insights and heatmaps	Not available	Not available
SLA-BASED ACCOUNTABILITY	Automated SLA tracking and escalation	Mostly manual	Not defined
RESPONSE TIME	Fast	Moderate	Slow